

Markscheme

Mathematics: analysis and approaches
Practice question bank

148. (a)

Given that $z_1 = 4 + 3i$.

Find $\frac{1}{z_1}$ in the form $a + bi$.

$$\begin{aligned}\frac{1}{z_1} &= \frac{\bar{z}_1}{|z_1|^2} = \frac{4 - 3i}{25} && (M1) \text{ A1} \\ &= 0.16 - 0.12i && \text{A1}\end{aligned}$$

[5 marks]

(b)

Hence find the modulus of $\frac{1}{z_1}$.

$$\begin{aligned}\left| \frac{1}{z_1} \right| &= \frac{1}{|z_1|} = \frac{1}{5} && (M1) \\ &= 0.2 && \text{A1}\end{aligned}$$

[2 marks]